

Research Proposal: Multimodal AI Systems

Executive Summary

This proposal outlines a 12-month research initiative to develop multimodal AI systems capable of jointly reasoning over text, images, and structured data. The project aims to produce novel architectures and benchmark datasets for evaluating cross-modal understanding.

Problem Statement

Current AI systems excel at individual modalities but struggle to integrate information across text and vision simultaneously. Real-world applications such as medical imaging analysis and document understanding require robust multimodal reasoning capabilities.

Methodology

We propose a three-phase approach: (1) develop a unified encoder architecture using cross-attention between modalities, (2) curate a large-scale multimodal dataset from publicly available sources, and (3) evaluate on established benchmarks including VQA, GQA, and our new DocQA suite.

Timeline and Budget

Phase 1 (months 1-4): Architecture design and baseline experiments. Phase 2 (months 5-8): Dataset curation and large-scale training. Phase 3 (months 9-12): Evaluation, ablations, and paper submission. Estimated budget: \$240,000 including compute costs and personnel.